Theory

ISPhO

A1-1

English

Ferromagnetic domain wall dynamics

Part A. Single magnetic moment (2.0 points)

A.1 (0.3pt)

A.2 (0.9pt)

$m_z(t) =$

A.3 (0.8pt)

$\Omega =$

Part B. Static domain wall (3.5 points)

B.1 (0.5pt)

(Formula) $B_{\mathrm{ex}} =$

(Value) $B_{\mathrm{ex}} =$

(Formula) $B_{\mathrm{an}} =$

(Value) $B_{\mathrm{an}} =$

(Formula) $\omega =$

(Value) $\omega =$

B.2 (1.0pt)

$\alpha =$

$T =$

$d =$

B.3 (1.5pt)

β	d

B.4 (0.5pt)

$c =$

Part C. Domain wall motion in external field (4.5 points)

C.1 (0.2pt)

Direction (to the right / to the left):

C.2 (2.0pt)

b	v	φ

C.3 (0.4pt)
Plot:



$dv/db =$

C.4 (0.6pt)
Plot:



$v_c =$ $b_{\mathrm{cr}} =$

C.5 (0.3pt)

$v_c =$ $B_{\mathrm{cr}} =$

C.6 (1.0pt)

t	x

Plot:

